

A Technical Introduction to Anomaly detection

Section 1

In data analysis, anomaly detection (also referred to as outlier detection and sometimes as novelty detection) is generally understood to be the identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well defined notion of normal behavior. Such examples may arouse suspicions of being generated by a different mechanism, or appear inconsistent with the remainder of that set of data. Anomaly detection finds application in many domains including cybersecurity, medicine, machine vision, statistics, neuroscience, law enforcement and financial fraud to name only a few. Anomalies were initially searched for clear rejection or omission from the data to aid statistical analysis, for example to compute the mean or standard deviation. They were also removed to better predictions from models such as linear regression, and more recently their removal aids the performance of machine learning algorithms. However, in many applications anomalies themselves are of interest and are the observations most desirous in the entire data set, which need to be identified and separated from noise or irrelevant outliers. Three broad categories of anomaly detection techniques exist. Supervised anomaly detection techniques require a data set that has been labeled as "normal" and "abnormal" and involves training a classifier. However, this approach is rarely used in anomaly detection due to the general unavailability of labelled data and the inherent unbalanced nature of the classes. Semi-supervised anomaly detection techniques assume that some portion of the data is labelled. This may be any combination of the normal or anomalous data, but more often than not, the techniques construct a model representing normal behavior from a given normal training data set, and then test the likelihood of a test instance to be generated by the model. Unsupervised anomaly detection techniques assume the data is unlabelled and are by far the most commonly used due to their wider and relevant application.

Section 2

The Subspace Outlier Degree (SOD) identifies attributes where a sample is normal, and attributes in which the sample deviates from the expected. Correlation Outlier Probabilities (COP) compute an error vector of how a sample point deviates from an expected location, which can be interpreted as a counterfactual explanation: the sample would be normal if it were moved to that location.

Section 3

Intrusion detection Anomaly detection was proposed for intrusion detection systems (IDS) by Dorothy Denning in

1986. Anomaly detection for IDS is normally accomplished with thresholds and statistics, but can also be done with soft computing, and inductive learning. Types of features proposed by 1999 included profiles of users, workstations, networks, remote hosts, groups of users, and programs based on frequencies, means, variances, covariances, and standard deviations. The counterpart of anomaly detection in intrusion detection is misuse detection.

Section 4

Software ELKI is an open-source Java data mining toolkit that contains several anomaly detection algorithms, as well as index acceleration for them. PyOD is an open-source Python library developed specifically for anomaly detection. scikit-learn is an open-source Python library that contains some algorithms for unsupervised anomaly detection. Wolfram Mathematica provides functionality for unsupervised anomaly detection across multiple data types

Section 5

Methods Many anomaly detection techniques have been proposed in literature. The performance of methods usually depend on the data sets. For example, some may be suited to detecting local outliers, while others global, and methods have little systematic advantages over another when compared across many data sets. Almost all algorithms also require the setting of non-intuitive parameters critical for performance, and usually unknown before application. Some of the popular techniques are mentioned below and are broken down into categories:

Section 6

IoT systems Anomaly detection is critical for the security and efficiency of Internet of Things (IoT) systems. It helps in identifying system failures and security breaches in complex networks of IoT devices. The methods must manage real-time data, diverse device types, and scale effectively. Garg et al. have introduced a multi-stage anomaly detection framework that improves upon traditional methods by incorporating spatial clustering, density-based clustering, and locality-sensitive hashing. This tailored approach is designed to better handle the vast and varied nature of IoT data, thereby enhancing security and operational reliability in smart infrastructure and industrial IoT systems.

Section 7

Preprocessing Preprocessing data to remove anomalies can be an important step in data analysis, and is done for a number of reasons. Statistics such as the mean and standard deviation are more accurate after the removal of anomalies, and the visualisation of data can also be improved. In supervised learning, removing the anomalous data from the dataset often results in a statistically significant increase in accuracy.

Section 8

Neural networks Replicator neural networks, autoencoders, variational autoencoders, long short-term memory neural networks Bayesian networks Hidden Markov models (HMMs) Minimum Covariance Determinant Deep Learning Convolutional Neural Networks (CNNs): CNNs have shown exceptional performance in the unsupervised learning domain for anomaly detection, especially in image and video data analysis. Their ability to automatically and hierarchically learn spatial hierarchies of features from low to high-level patterns makes them particularly suited for detecting visual anomalies. For instance, CNNs can be trained on image datasets to identify atypical patterns indicative of defects or out-of-norm conditions in industrial quality control scenarios. Simple Recurrent Units (SRUs): In time-series data, SRUs, a type of recurrent neural network, have been effectively used for anomaly detection by capturing temporal dependencies and sequence anomalies. Unlike traditional RNNs, SRUs are designed to be faster and more parallelizable, offering a better fit for real-time anomaly detection in complex systems such as dynamic financial markets or predictive maintenance in machinery, where identifying temporal irregularities promptly is crucial. Foundation models: Since the advent of large-scale foundation models that have been used successfully on most downstream tasks, they have also been adapted for use in anomaly detection and segmentation. Methods utilizing pretrained foundation models include using the alignment of image and text embeddings (CLIP, etc.) for anomaly localization, while others may use the inpainting ability of generative image models for reconstruction-error based anomaly detection.

Section 9

Quantum Quantum machine learning approaches have been investigated for anomaly detection on near-term quantum hardware. Quantum support vector machines (QSVMs) using quantum kernel methods, such as the ZZFeatureMap, have been applied to classification of normal versus anomalous instances by leveraging high-dimensional quantum feature spaces.